In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sbn
```

In [2]:
```python
# Loading dataset
df = pd.read_csv('train.csv')
df.head()
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	(
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

In [3]:
```python
# Provides Basic info about columns, types, and missing values
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [4]:
```python
#Provides Summary statistics for numerical columns
df.describe()
```

Out[4]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

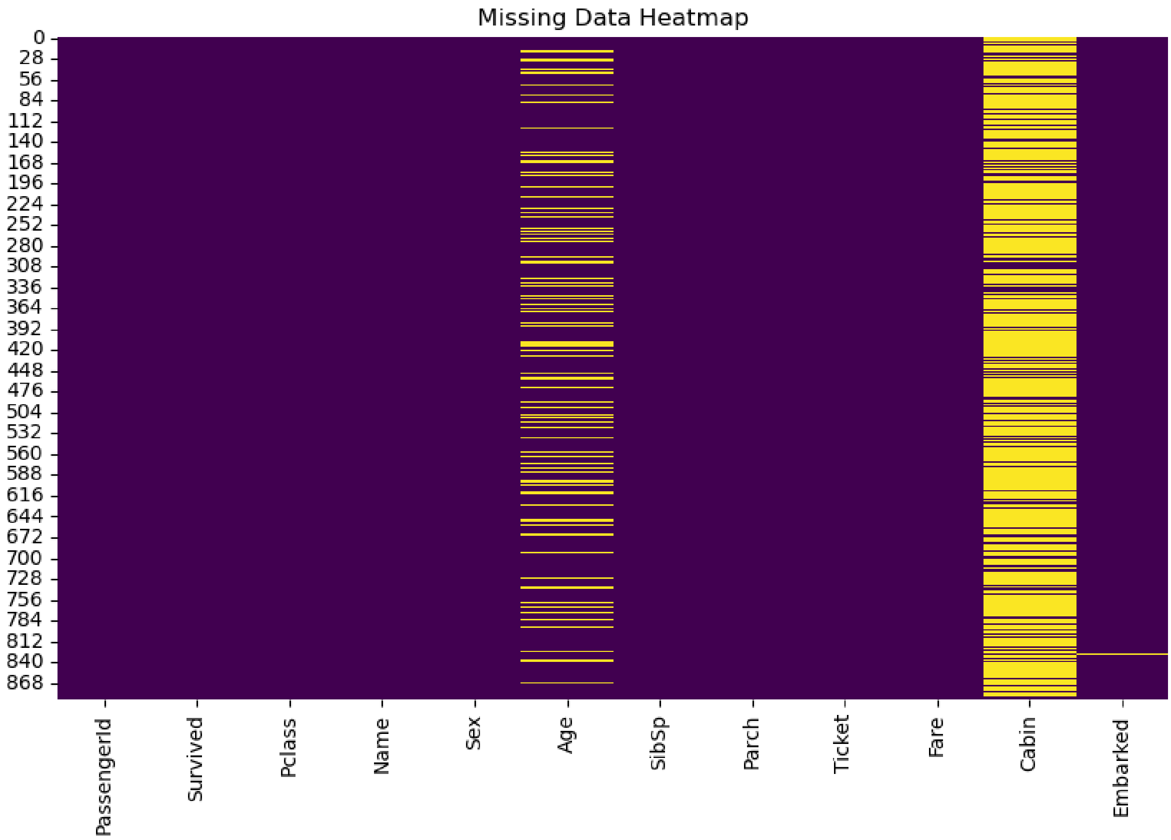
In [5]:
```python
# Sum of null values in each column
df.isnull().sum()
```

Out[5]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [6]:
```python
# Count of unique values for categorical variables
df['Sex'].value_counts()
df['Embarked'].value_counts()
df['Pclass'].value_counts()
```

Out[6]:
```
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

In [7]:
```python
#Visualizing Missing Data

plt.figure(figsize=(10,6))
sbn.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.title('Missing Data Heatmap')
plt.show()
```



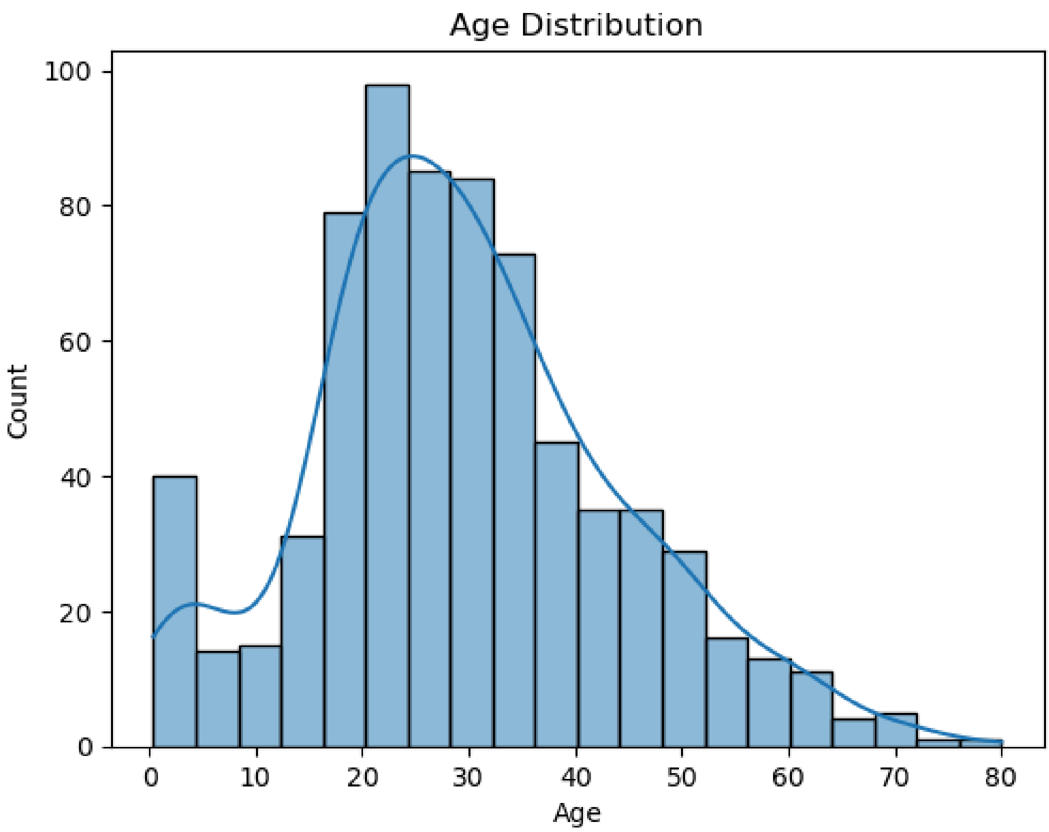
In [8]:

```
#Feature Engineering: Create New Features

# Creating a new feature: FamilySize
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1

# Log transformation for Fare to handle skew
df['Fare_log'] = np.log1p(df['Fare'])
```

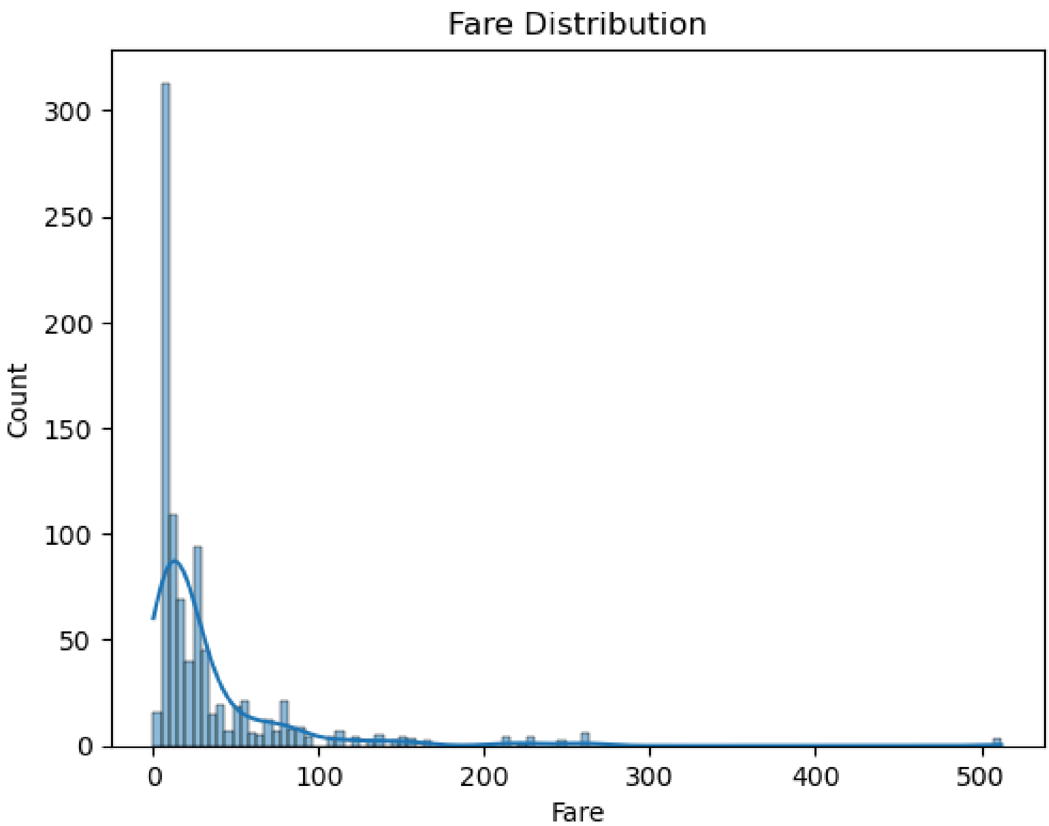
Univariate Analysis

In [9]:

```
#Histogram to show Distribution of Age
sbn.histplot(df['Age'].dropna(), kde=True)
plt.title('Age Distribution')
plt.show()
```

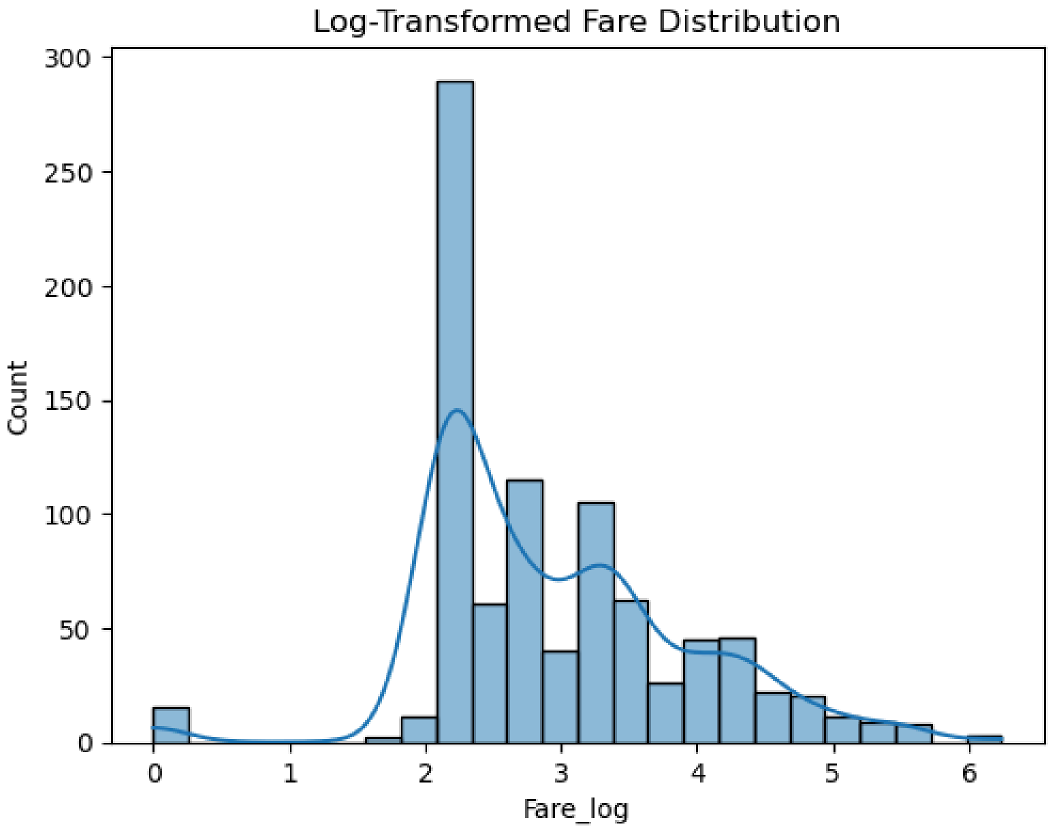


In [10]:
```
# Fare Distribution
sbn.histplot(df['Fare'], kde=True)
plt.title('Fare Distribution')
plt.show()
```



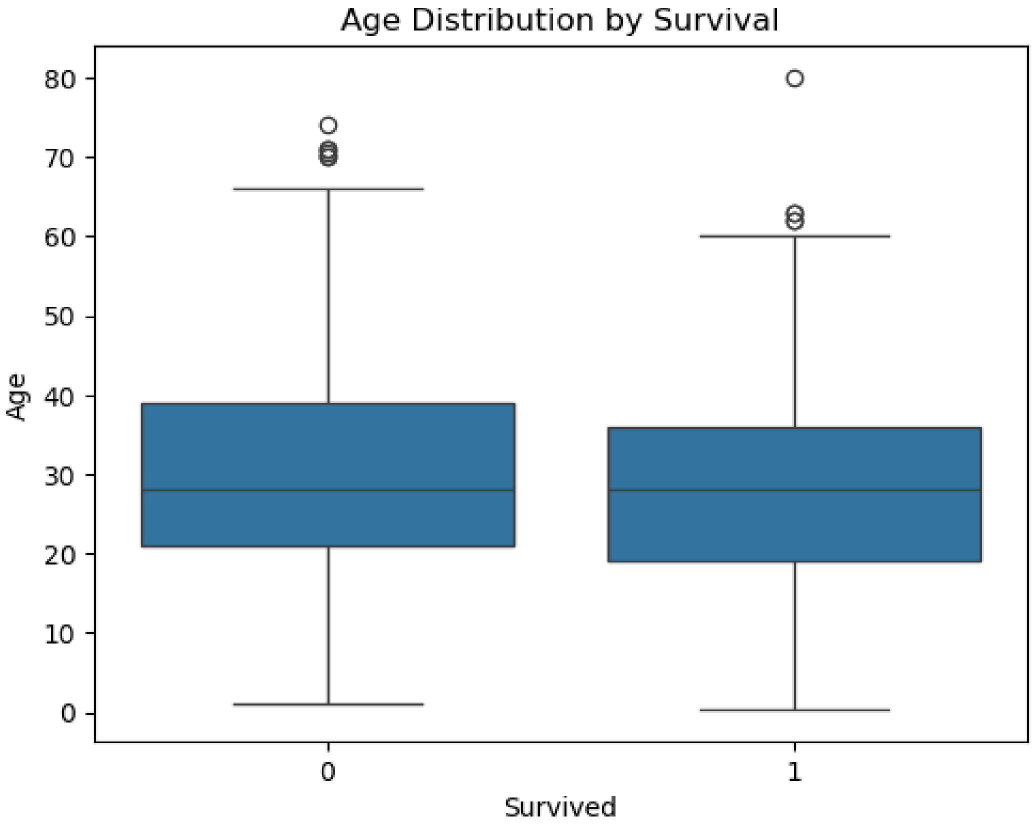
In [11]:
```
# Log-transformed Fare
sbn.histplot(df['Fare_log'], kde=True)
```

```
plt.title('Log-Transformed Fare Distribution')
plt.show()
```

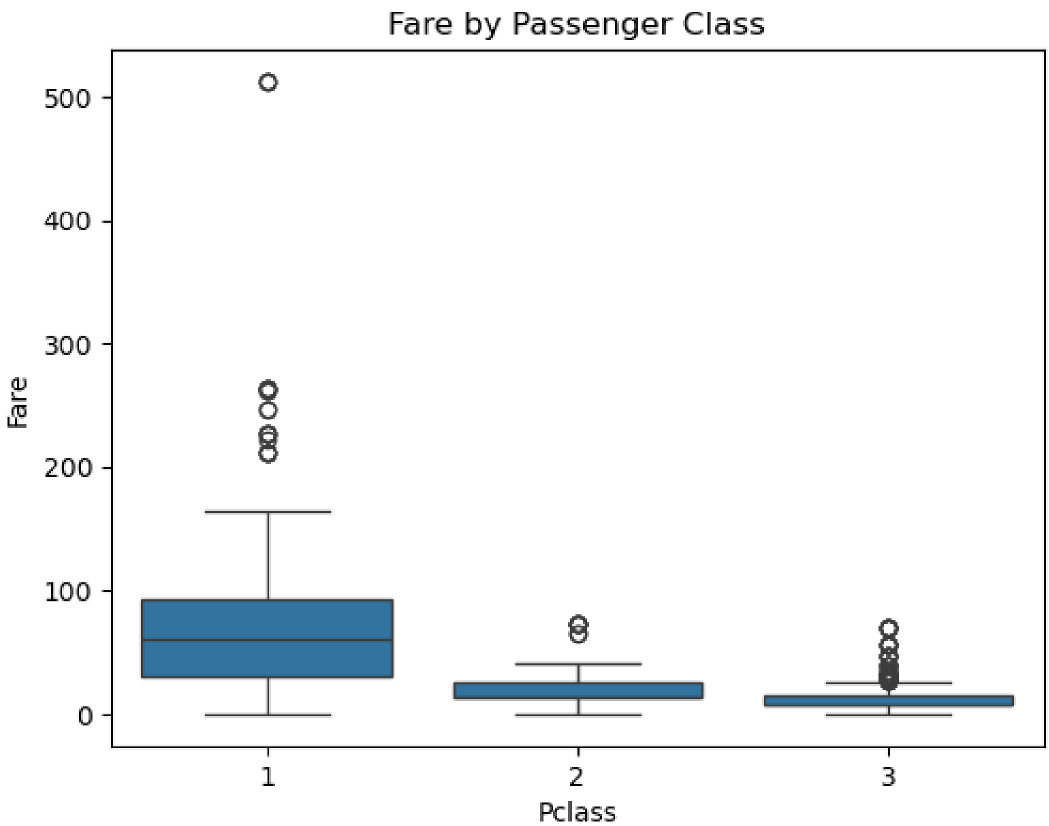


Bivariate Analysis

In [12]:
```
#Boxplot - Age vs Survived

sbn.boxplot(x='Survived', y='Age', data=df)
plt.title('Age Distribution by Survival')
plt.show()
```

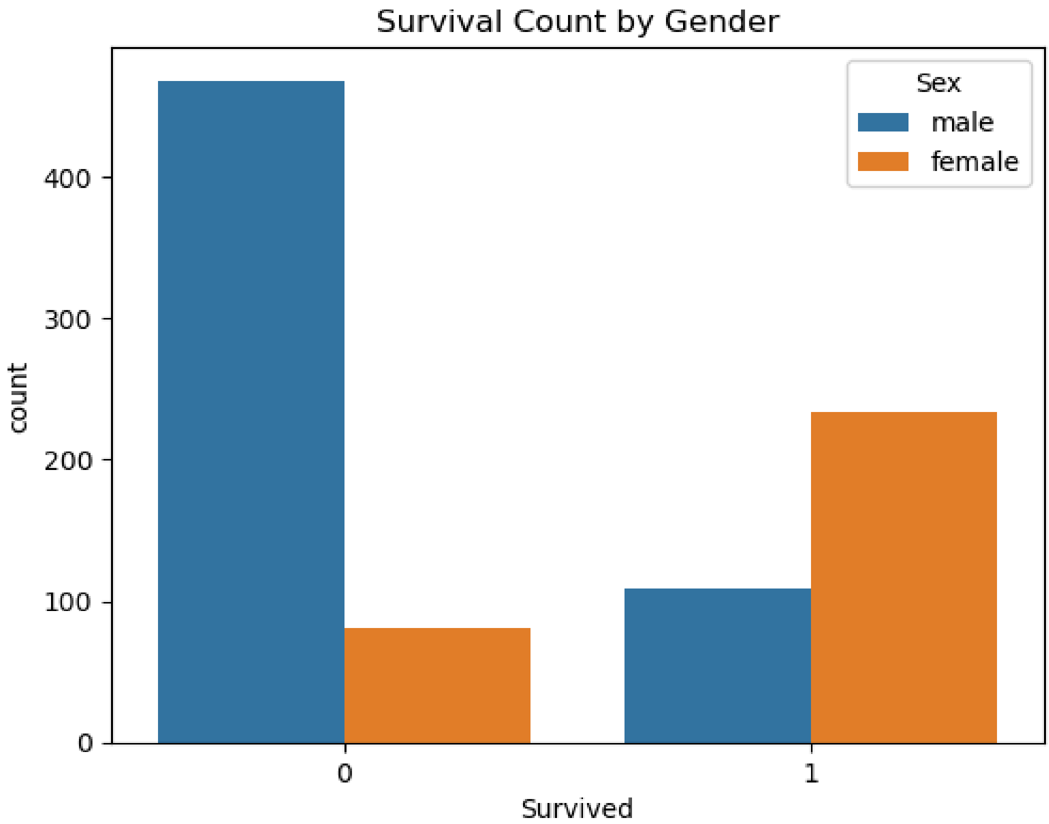


In [13]:
```
#Fare by Class
sbn.boxplot(x='Pclass', y='Fare', data=df)
plt.title('Fare by Passenger Class')
plt.show()
```



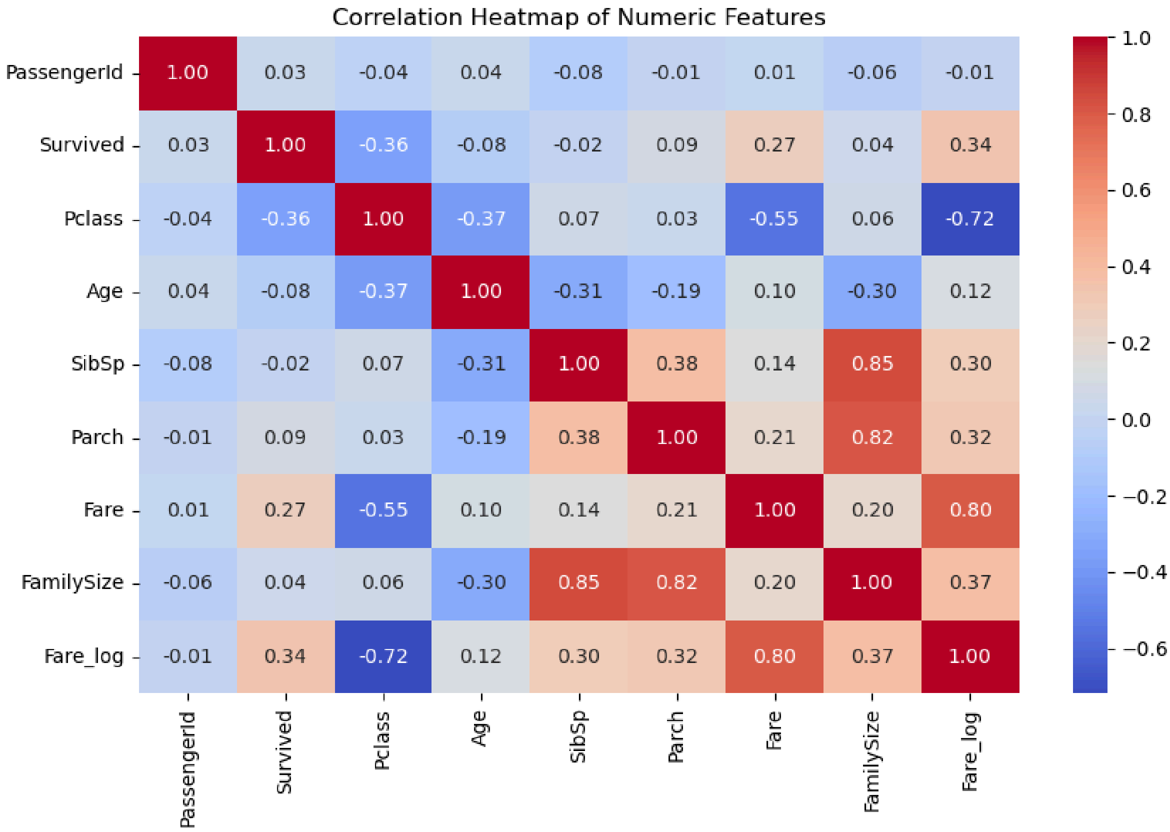
In [14]:
```
# Survival Count by Gender
sbn.countplot(x='Survived', hue='Sex', data=df)
```

```
plt.title('Survival Count by Gender')
plt.show()
```

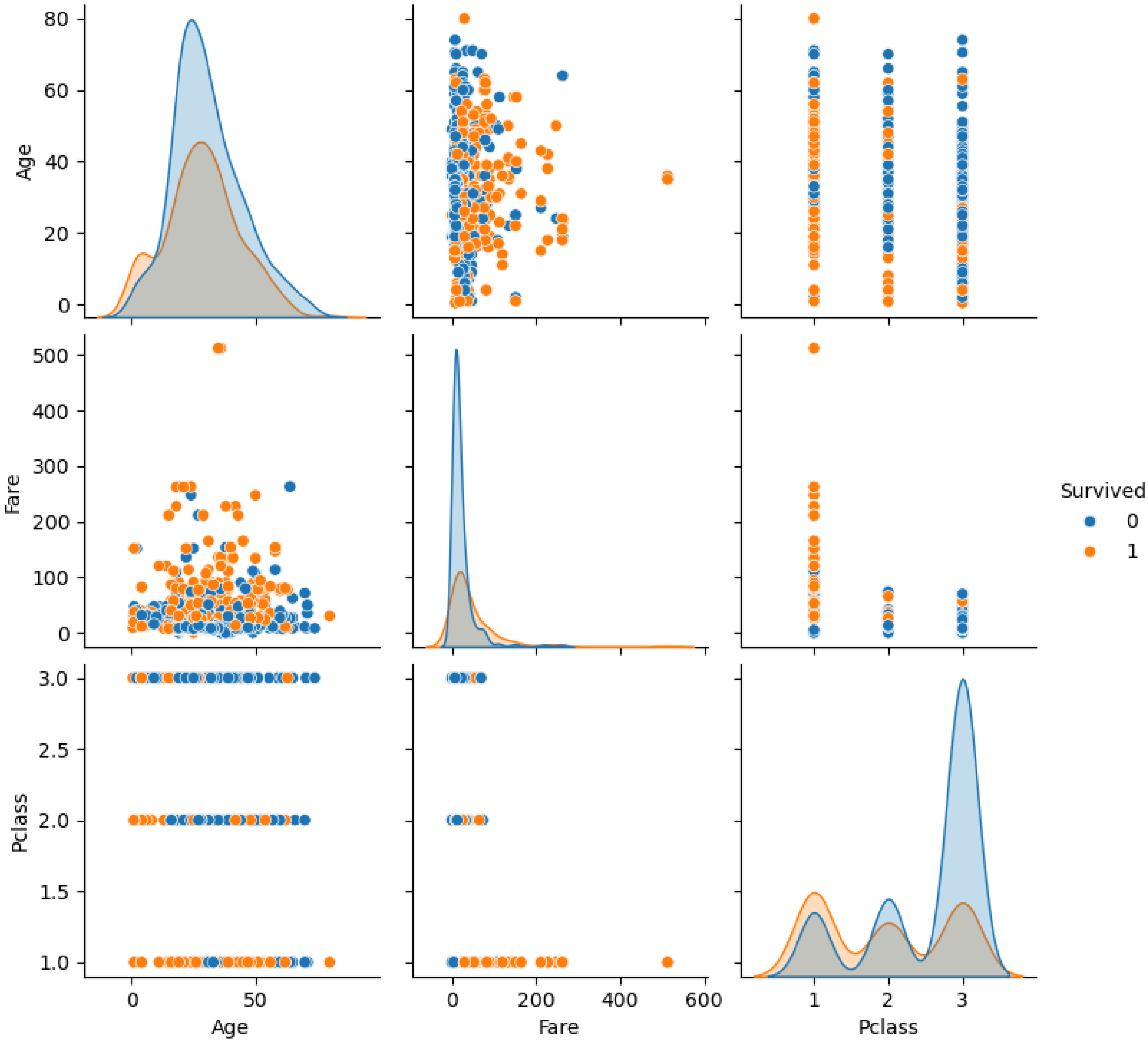


Heatmap – Correlation Matrix

In [15]:
```
# Filter only numeric columns and drop missing values
numeric_df = df.select_dtypes(include=['int64', 'float64']).dropna()

# Plot heatmap
plt.figure(figsize=(10, 6))
sbn.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap of Numeric Features')
plt.show()
```



In [16]:
```
#Pairplot - Visualize relationships

sbn.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']], hue='Survived')
plt.show()
```



In [17]:

```
# Grouped Survival Rates

print("Survival Rate by Class:\n", df.groupby('Pclass')['Survived'].mean())
print("Survival Rate by Sex:\n", df.groupby('Sex')['Survived'].mean())
print("Survival Rate by Embarked:\n", df.groupby('Embarked')['Survived'].mean())
print("Survival Rate by Family Size:\n", df.groupby('FamilySize')['Survived'].mean())
```

```
Survival Rate by Class:
 Pclass
1    0.629630
2    0.472826
3    0.242363
Name: Survived, dtype: float64
Survival Rate by Sex:
 Sex
female    0.742038
male      0.188908
Name: Survived, dtype: float64
Survival Rate by Embarked:
 Embarked
C    0.553571
Q    0.389610
S    0.336957
Name: Survived, dtype: float64
Survival Rate by Family Size:
 FamilySize
1     0.303538
2     0.552795
3     0.578431
4     0.724138
5     0.200000
6     0.136364
7     0.333333
8     0.000000
11    0.000000
Name: Survived, dtype: float64
```

In [ ]: